MOHAMED MUFASSIR RISWAN

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PROFESSIONAL SUMMARY

3rd-year Computer Systems Engineering undergraduate (SLIIT) with a strong foundation in embedded systems, real-time control, and robotics. Picks up new tools and frameworks quickly — from AVR assembly to ROS 2 — and applies them to build working hardware-software systems. Combines hands-on project experience across IoT, control systems, and full-stack development with freelance client delivery under MR Solutions.

EDUCATION

BSc (Hons) in Computer Systems Engineering *Nov 2023 – Expected Nov 2027*

Sri Lanka Institute of Information Technology (SLIIT)

Relevant Modules: Real-Time Systems, Digital Logic Design, Control Theory, Computer Architecture, Data Structures & Algorithms, Computer Networks

EXPERIENCE

Software Engineering Intern — ZILLIT (Startup), Nittambuwa *Nov 2025 – Feb 2026*

- Led sole development of a cross-platform Point of Sale (POS) desktop application using Electron.js and TypeScript, built as part of a broader IoT ecosystem spanning desktop and mobile
- Built the front-end of an enterprise-level financial management system in Vue.js, integrating with a Java Spring Boot backend
- Worked across React, TypeScript, Vue.js, Electron.js, PostgreSQL, and Docker in a fast-paced startup environment, collaborating on system architecture decisions

TECHNICAL SKILLS

Embedded & Hardware: AVR (ATmega), Arduino, ESP32, Raspberry Pi, UART/SPI/I2C, Sensors & Actuators, PCB Design, Oscilloscope & Lab Instruments

Languages: C/C++, Embedded C, AVR Assembly, Python, JavaScript, TypeScript, Rust (Basic)

Robotics & Simulation: Fusion 360, Proteus, ROS 2 & Gazebo (in progress), URDF/robot modeling

Software Stack: React, Vue.js, Node.js/Express, Spring Boot (Java),Fast API (Python), OpenCV, Media Pipe, PostgreSQL/MySQL, Docker

Tools: Git, Linux, Arduino IDE, MATLAB, VS Code, EasyEDA

PROJECTS

AquaLoRa — LoRa-Based IoT Water Quality Monitoring for Fish Farms *2026*

ESP32 + LoRa, Electron/React, FastAPI/PostgreSQL

- Owned the Dashboard & Database subsystem of a 4-person solar-powered IoT system, achieving confirmed 1km-range LoRa communication between sensor and master nodes
- Built a cross-platform desktop app (Electron + React + TypeScript) with local SQLite storage and Recharts visualization, backed by a FastAPI + PostgreSQL cloud sync layer
- Designed a rule-based Fish Health Index engine (temperature, turbidity, TDS) running as a Python subprocess to flag Normal/Warning/Critical conditions in real time

Stack: ESP32, LoRa (SX1278), Embedded C, Electron, React, TypeScript, Python, FastAPI, PostgreSQL

Hand Gesture Controlled Robot – using OpenCV *2025*

ESP32 + Python , OpenCV/Desktop App,WiFi

- A Dashboard that connected to OpenCV Library that identify the Hand Gestures(MediaPipe) and Send Commands.
- ESP32 MCU Controlled DIY Car kit that execute those commands, so the car move accordingly (Connection Through WiFi).

Stack: ESP32, Embedded C, Python, OpenCV, MediaPipe, WiFi

ESP32-CAM Remote-Controlled Car with Live Video Streaming *2025*

- Built a WiFi-controlled RC car with live video streaming, using the ESP32-CAM module as both access point and video server
- Developed a mobile app and web-based controller (live on Vercel) for real-time driving and camera feed viewing from any device on the network

  Stack: ESP32-CAM, Embedded C, JavaScript, Vercel, WiFi/HTTP streaming

Full-Stack Web Applications — Freelance Client Work (MR Solutions) *2025 – 2026*

- **Travel Ready Tours:** full-stack fleet management and booking system — React/TypeScript/Tailwind frontend, Node.js/Express/PostgreSQL/Prisma backend, Docker deployment, plus an Expo/React Native driver app with Firebase push notifications
- **Hutch Way Point:** metadata-driven workflow/project-management system with a configurable engine where Solution/Stage/Field/Transition tables drive runtime-rendered forms and json-logic rules
- Delivered several freelance business websites (React/Vite/TypeScript/Tailwind) for local clients

  Stack: React, TypeScript, Node.js/Express, PostgreSQL, Prisma, Docker, React Native/Expo

PID-Based DC Motor Speed Control System (Raspberry Pi + Python) *2026*

- Designed a real-time closed-loop control system using a PID algorithm with encoder feedback for precise RPM measurement and regulation
- Extended the project with a Python desktop GUI to visualize live motor behavior, RPM graphs, and PID tuning parameters

  Stack: Python, Raspberry Pi, PID control, encoder interfacing, data visualization

Other Notable Projects

- **AVR Autonomous Line-Following Car with Smart Parking (2026):** FSM-based robot with 3 IR sensors for line tracking and autonomous parking-bay detection via crossline logic
- **IoT Smart Surveillance & Access Control, ESP32 (2024):** camera-integrated Telegram bot system for real-time intrusion alerts and remote door lock/unlock over UART/I2C
- **3D CAD Design of a 6-DOF Robotic Arm, Fusion 360 (2026):** modeled full arm assembly with joint constraints and motion limits; exported for ROS/URDF integration

CERTIFICATIONS & LEARNING

- **Mastering ROS 2 for Robotics Programming Specialization** — Packt (Coursera) — In Progress
- Introduction to Embedded Systems Software and Development Environments — University of Colorado Boulder (Coursera)
- The Complete Full-Stack Web Developer Bootcamp — Dr. Angela Yu, Udemy

INTEREST AREAS

Embedded Systems | IoT | Robotics & Control Systems | Network Engineering | Telecommunications | Industrial Electronics | Software Engineering

REFERENCES

Prof. Anuradha Jayakody

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